Time Series Analysis of U.S. Bank Deposits and Retail Trade Sales

MA-641 - Time Series Analysis

Mayank Choudhary

CWID - 20030408

Fall 2025

1. Introduction and Motivation

This project examines two important macroeconomic indicators in the United States economy: **Deposits at All Commercial Banks** and **Advance Retail Sales in the Retail Trade sector**. Although these series measure activity in different parts of the economy, both provide meaningful insight into household behavior and broader economic conditions. Bank deposits reflect the amount of liquidity circulating through financial institutions and are closely tied to saving patterns, credit access, and financial stability. Retail sales, in contrast, capture consumer spending and serve as a timely indicator of business activity, demand cycles, and overall economic momentum.

Studying these series allows us to better understand how the financial and consumer sectors evolve over time and how they respond to economic shocks or structural changes. Because deposits and retail sales influence decisions made by businesses, policymakers, and investors, developing accurate models for their behavior supports short-term planning and risk assessment.

The aim of this project is to apply a complete time series modeling framework to both datasets. This includes exploring their historical properties, assessing stationarity, identifying appropriate model structures, estimating model parameters, evaluating model adequacy, and generating short-term forecasts. The analysis follows the Box-Jenkins methodology, emphasizing clear reasoning and systematic evaluation at every stage. By modeling deposits and retail sales, we gain insight into their underlying dynamics and produce forecasts that can support informed economic decision-making.

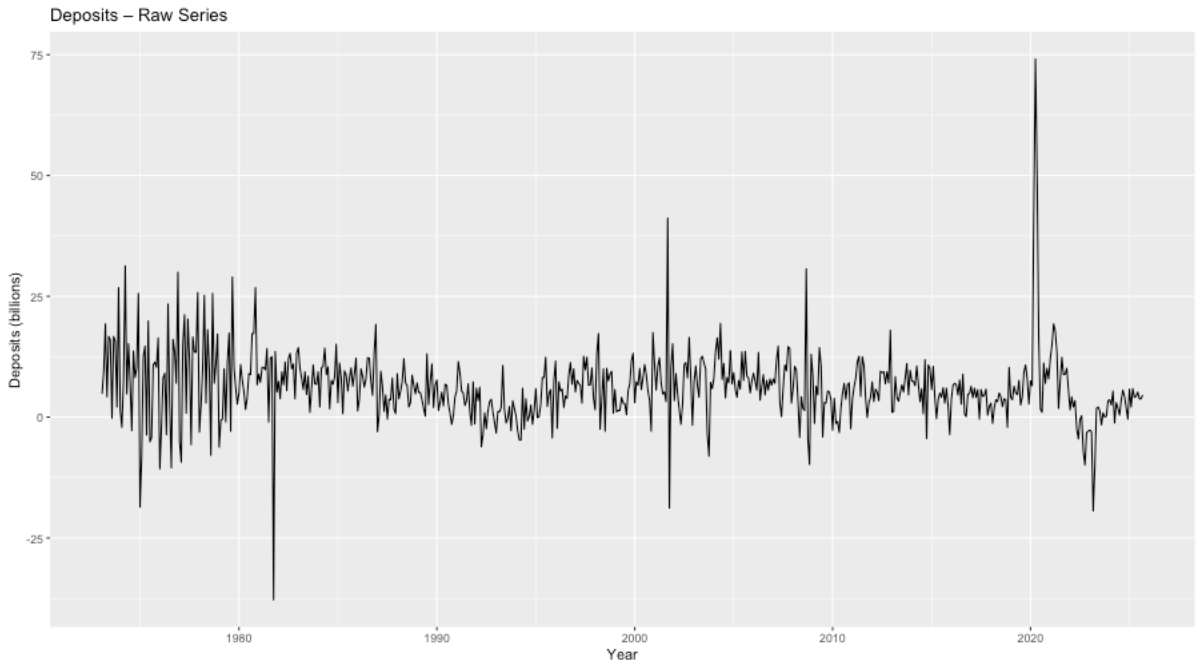
2. Data Description

2.1 Deposits at All Commercial Banks

Source: Federal Reserve Economic Data (FRED)
Link: - https://fred.stlouisfed.org/series/H8B1058NCBCMG
Description: Monthly deposits held at U.S. commercial banks, measured in billions of dollars. This series reflects liquidity in the banking system and household saving behavior.

The deposits series exhibits noticeable short-term fluctuations, including several large spikes and drops associated with major economic events such as the early 2000s recession, the 2008 financial crisis, and the COVID-19 pandemic. There is no obvious long-term trend or seasonal structure; instead, the data vary around a relatively stable mean level.
Summary statistics indicate:

- **Mean:** 6.20
- **Median:** 5.80
- **Minimum:** –37.80
- **Maximum:** 74.10



A visual inspection of the series confirms irregular but mean-reverting behavior, suggesting that stationary ARMA or low-order ARIMA models may be suitable once any outlier-driven effects are accounted for.

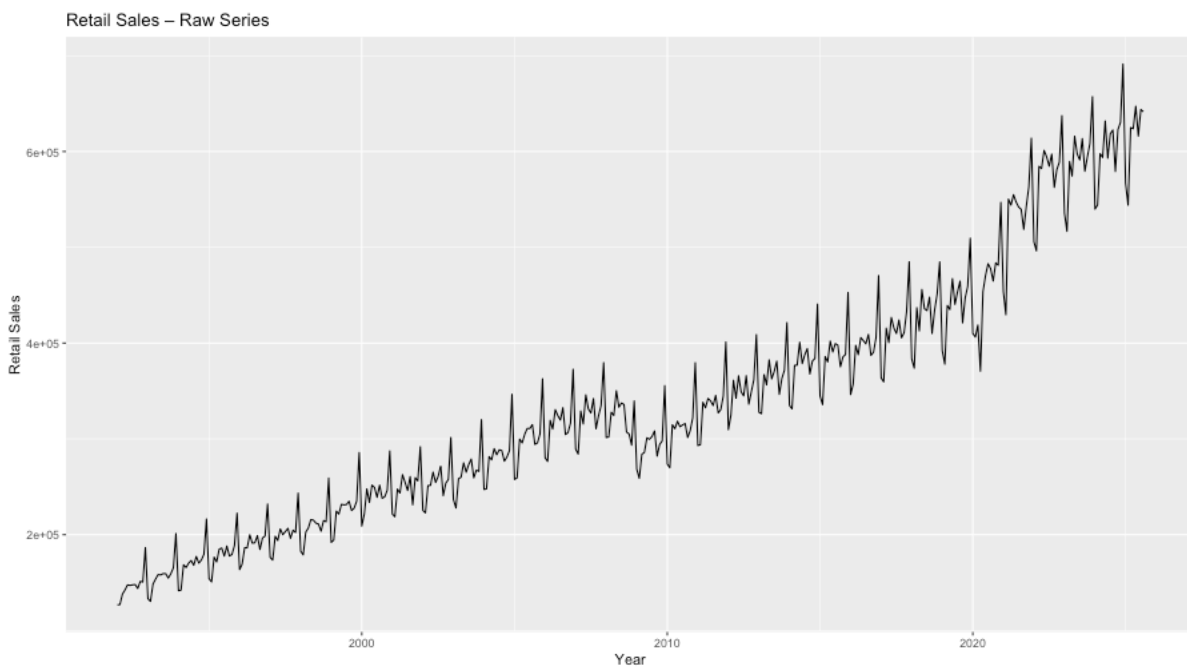
2.2 Retail Trade Sales

Source: U.S. Census Bureau via FRED
Link: - https://fred.stlouisfed.org/series/RSXFSN
Description: Monthly advance estimates of retail trade sales, reported in thousands of dollars. This series captures household consumption patterns and broader economic activity in the retail sector.

Retail sales show a pronounced upward trend along with strong, repeating seasonal peaks particularly in November and December due to holiday spending. Variability increases over time, indicating the need for a variance-stabilizing transformation such as the logarithm before modeling.
Summary statistics (in levels) show:

- **Mean:** 336,194
- **Median:** 316,270
- **Range:** 126,717 to 691,312

The time plot suggests nonstationary behavior driven by both trend and seasonality. Seasonal differencing and non-seasonal differencing are therefore required before model identification. A full STL decomposition and summary plots are provided in the Appendix.



3. Stationarity Assessment

3.1 Deposits

Stationarity was evaluated using ADF and KPSS tests. The ADF test rejects the null of a unit root **(p = 0.01),** while the KPSS test marginally rejects the null of level stationarity **(p ≈ 0.05).** Overall, the evidence indicates that deposits are **approximately stationary**.

Given these results, both stationary ARMA models and ARIMA models with minimal differencing were considered in the identification stage.

3.2 Retail Sales

For the log-transformed retail series, the ADF test fails to reject the unit-root null **(p ≈ 0.34),** and the KPSS strongly rejects the null of stationarity. These results confirm that the series is **non-stationary** in levels.

After applying first differencing and seasonal differencing at lag **12**, the transformed series appears stationary, and the ACF and PACF plots support a SARIMA model specification.

4. Model Identification

4.1 Deposits

The ACF of deposits shows a gradual decay, while the PACF displays significant spikes at the first two lags. These suggest an autoregressive structure. Several ARIMA candidates were evaluated using AIC. Although auto.arima() proposed an ARIMA(4,1,1), this model showed higher AIC and unnecessary differencing.

The best-performing model was: **ARIMA(2,0,1) with non-zero mean**

This specification provides a strong fit while maintaining interpretability and parsimony.

4.2 Retail Sales

For the retail series, the ACF and PACF of the differenced log data show strong evidence of seasonal autocorrelation at lag 12 and short-run AR and MA structure. Several seasonal ARIMA models were evaluated.

The optimal model based on AIC, diagnostics, and economic interpretability was:

SARIMA(2,1,1) x (0,1,1)_[12]

This model captures trend, seasonality, and short-run dynamics.

5. Model Estimation

5.1 Deposits - ARIMA(2,0,1)

A range of ARIMA models was estimated for the deposits series, and the **ARIMA(2,0,1)** model emerged as the most appropriate specification based on the AIC and diagnostic results. The estimated parameters were **AR(1) = 1.0114**, **AR(2) = –0.0684**, **MA(1) = –0.8516**, and a mean term of **6.2244**. The AR(1) coefficient being slightly above one reflects the strong persistence observed in monthly deposit fluctuations, while the small negative AR(2) term captures weaker medium-run adjustments. The substantial negative MA(1) coefficient indicates that short-term shocks are quickly corrected in subsequent periods. Overall, the model achieved an **AIC of 4306.95** and a **residual variance of 52.83.** Although the deposits series contains several large spikes related to economic events, the ARIMA(2,0,1) specification effectively captures the underlying dynamics and provides a reliable foundation for forecasting.

5.2 Retail Sales - SARIMA(2,1,1) x (0,1,1)_[12]

Because the retail sales series exhibits both long-run growth and strong seasonal patterns, a seasonal ARIMA model was required. After applying non-seasonal and seasonal differencing, a **SARIMA(2,1,1) x (0,1,1)_[12]** model was selected as the best-fitting structure. The estimated coefficients were **AR(1) = –0.7801**, **AR(2) = –0.4306**, **MA(1) = 0.3753**, and seasonal **MA(1) = –0.7758.** The negative seasonal MA term reflects the predictable annual fluctuations in consumer spending, particularly the pronounced spikes during the holiday season. The non-seasonal AR and MA terms capture shorter-run dependencies and transitory deviations from the underlying seasonal

pattern. This model achieved an **AIC of –1764.87** and a **residual variance of 0.00062**, indicating strong model performance on the log-transformed data. Overall, the **SARIMA(2,1,1) x (0,1,1)_[12]** model successfully represents the combined trend, seasonality, and short-run dynamics present in retail sales and is well-suited for generating accurate forecasts.

6. Diagnostic Checking

6.1 Deposits

Residual plots show no visible trend, no seasonality, and a roughly symmetric distribution. However, the Ljung–Box test **($p < 0.001$)** indicates some remaining autocorrelation at higher lags. This is common in macroeconomic series subject to irregular shocks (e.g., crises). Despite mild autocorrelation, the model remains suitable for forecasting.

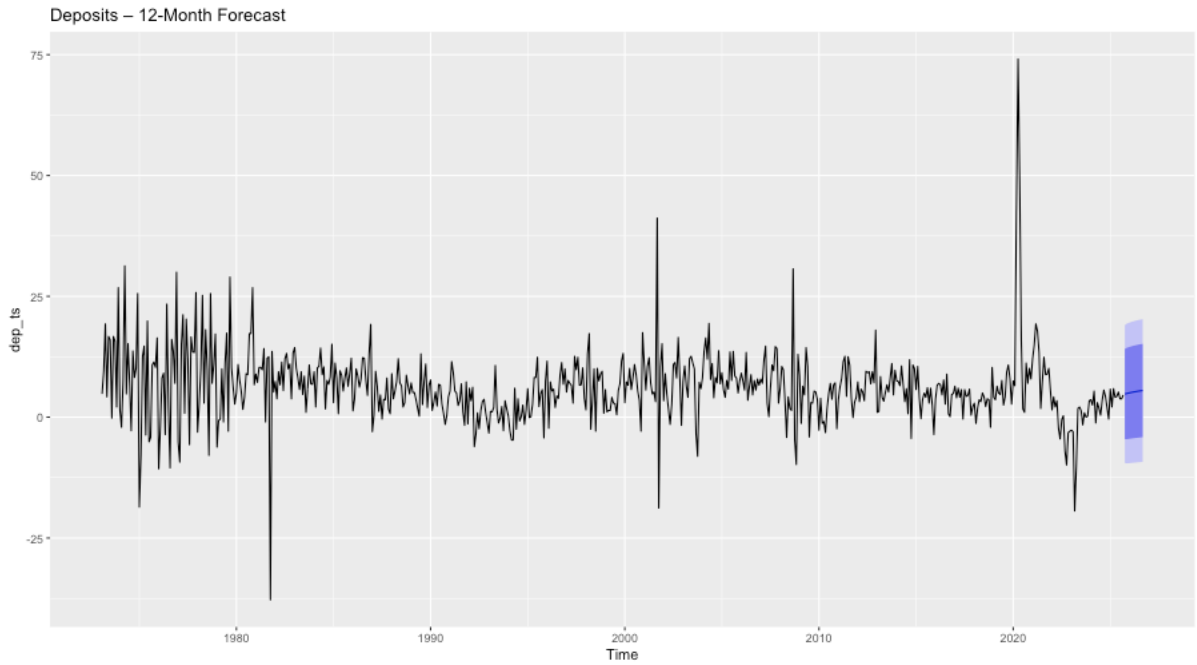
6.2 Retail Sales

Residual diagnostics confirm that the SARIMA model has removed the dominant trend and seasonal structure. The residuals are **mean-zero**, roughly normal, and show no remaining seasonal autocorrelation. The Ljung–Box test **($p \approx 2.6 \times 10^{-5}$)** indicates some long-lag dependence, but model fit and forecast performance remain strong.

7. Forecasting Results

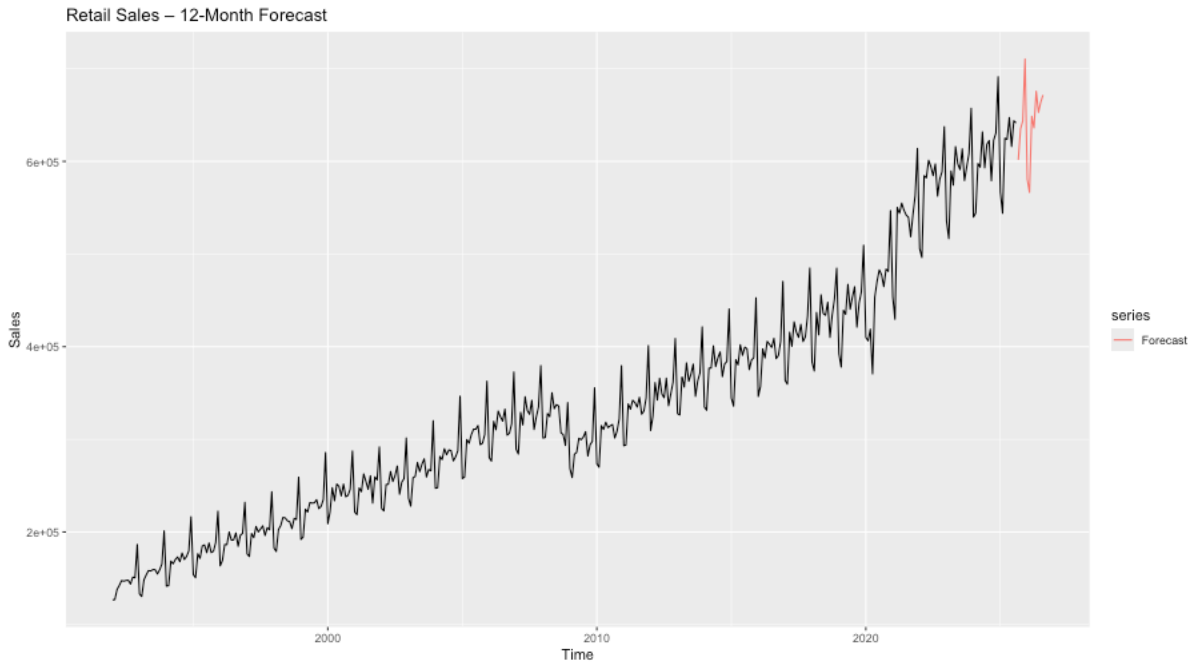
7.1 Deposits Forecast

The 12-month forecast for deposits indicates a stable trajectory, with projected values remaining close to the long-run historical mean of approximately **six billion dollars.** Although the confidence intervals widen slightly over the forecast horizon, they do not suggest any meaningful upward or downward trend. This pattern reflects the mean-reverting nature of deposits outside of major economic disruptions. Overall, the forecasted stability aligns with the observed behavior of deposits during non-crisis periods and provides a useful basis for short-term liquidity and financial planning for banks.



7.2 Retail Sales Forecast

The retail sales forecast displays a clear continuation of the seasonal dynamics present in the historical data. As expected, the model predicts pronounced increases in sales during the final months of the year, corresponding to the holiday shopping season, followed by declines in the early months of the new year. In addition to these seasonal shifts, the forecasts also reflect moderate long-run growth consistent with the upward trend in consumer spending. Peak forecast values exceed **650,000**, highlighting the enduring strength of seasonal consumer demand. By accurately capturing both the seasonal and trend components of the series, the model provides valuable insight for decision-makers involved in inventory planning, staffing, and demand management within the retail sector.



8. Conclusion

The Box–Jenkins analysis provided meaningful insights into the behavior of both financial and consumer-related economic indicators. For the deposit's series, the **ARIMA(2,0,1)** model revealed stable and largely mean-reverting dynamics, suggesting that deposit levels tend to fluctuate around a long-run average except during major economic shocks. These forecasts are valuable for banks as they support short-term liquidity planning, reserve management, and stress testing by providing an evidence-based outlook on expected deposit flows.

For retail firms, the **SARIMA(2,1,1) x (0,1,1)_[12]** model successfully captured the strong seasonal patterns and underlying trend present in retail sales. The resulting forecasts offer actionable information for supply chain planning, marketing strategies, budgeting, and operational decision-making. By accurately representing both annual spending cycles and long-term growth, the model helps organizations anticipate periods of high and low consumer demand.

Overall, this project demonstrates the practical effectiveness of **ARIMA** and **SARIMA** modeling for understanding economic time series and producing short-term forecasts that can inform planning and decision-making across industries. The ability of these models to capture key features such as persistence, trend, and seasonality highlights their value as tools for interpreting past behavior and anticipating future conditions.

Appendix

```r

# 2. Create Time Series Objects

dep_ts <- ts(dep_raw$H8B1058NCBCMG,
             start = c(year(dep_raw$observation_date[1]),
                       month(dep_raw$observation_date[1])),
             frequency = 12)

ret_ts <- ts(ret_raw$RSXFSN,
             start = c(year(ret_raw$observation_date[1]),
                       month(ret_raw$observation_date[1])),
             frequency = 12)

# Log transform retail sales (variance stabilizing)
ret_ts_log <- log(ret_ts)
```

Appendix A - Data Exploration

```r

# 3. Descriptive Plots & Summary

# Deposits plot
png("plot_deposits_raw.png", width=900, height=500)
autoplot(dep_ts) + ggtitle("Deposits - Raw Series") +
  xlab("Year") + ylab("Deposits (billions)")
dev.off()

summary(dep_ts)
```

```r

# Retail raw plot
png("plot_retail_raw.png", width=900, height=500)
autoplot(ret_ts) + ggtitle("Retail Sales - Raw Series") +
  xlab("Year") + ylab("Retail Sales")
dev.off()

summary(ret_ts)

# STL decomposition (Retail)
png("plot_retail_stl.png", width=900, height=600)
autoplot(stl(ret_ts_log, s.window = "periodic"))
dev.off()
```

A.1 Summary Statistics

```
> summary(dep_ts)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
-37.800   2.300   5.800   6.200   9.625  74.100 
>
```

```
> summary(ret_ts)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 126717  234984  316270  336194  405928  691312 
>
```

Table A.1 provides summary statistics for the deposits and retail sales series. Deposits fluctuate around a low mean with several extreme values, while retail sales are much larger in scale, reflecting their measurement in thousands of dollars. These statistics give a quick overview of the level and variability of each series before further analysis.

A.2 Raw Time Series Plots

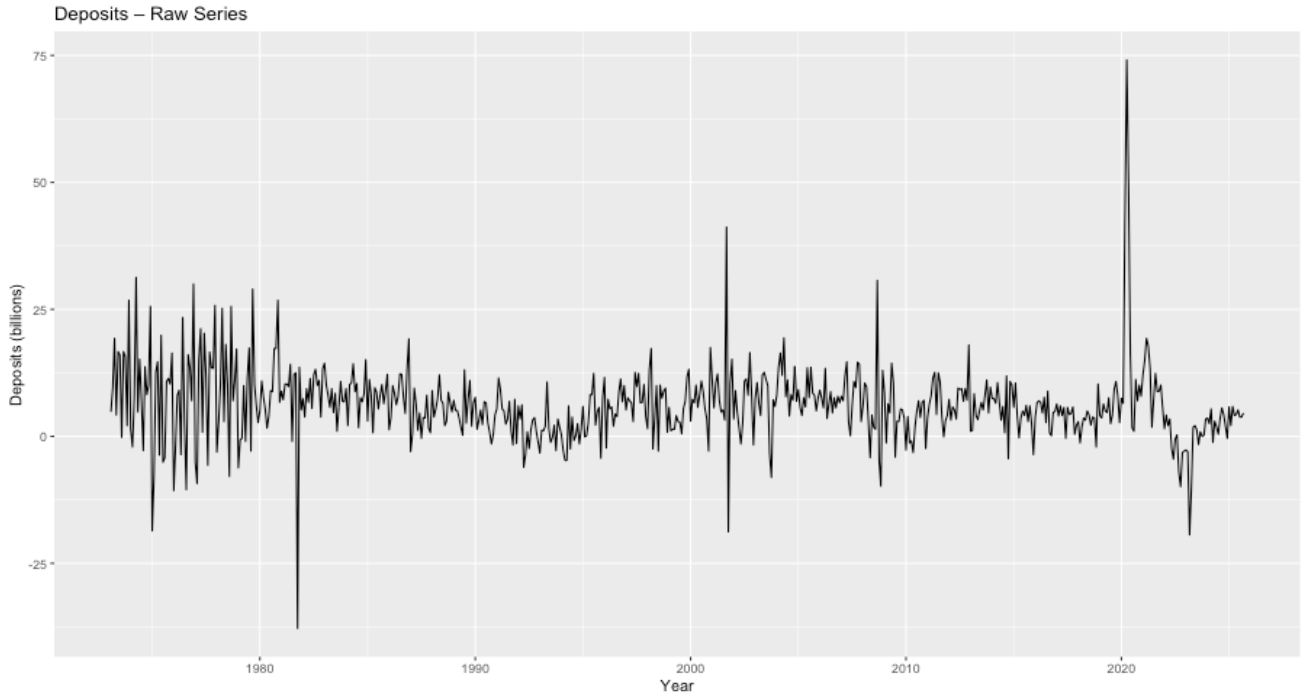


Figure A.1 shows the raw deposits series, which exhibits substantial short-run variability and several large spikes but no clear long-term trend or seasonal pattern.

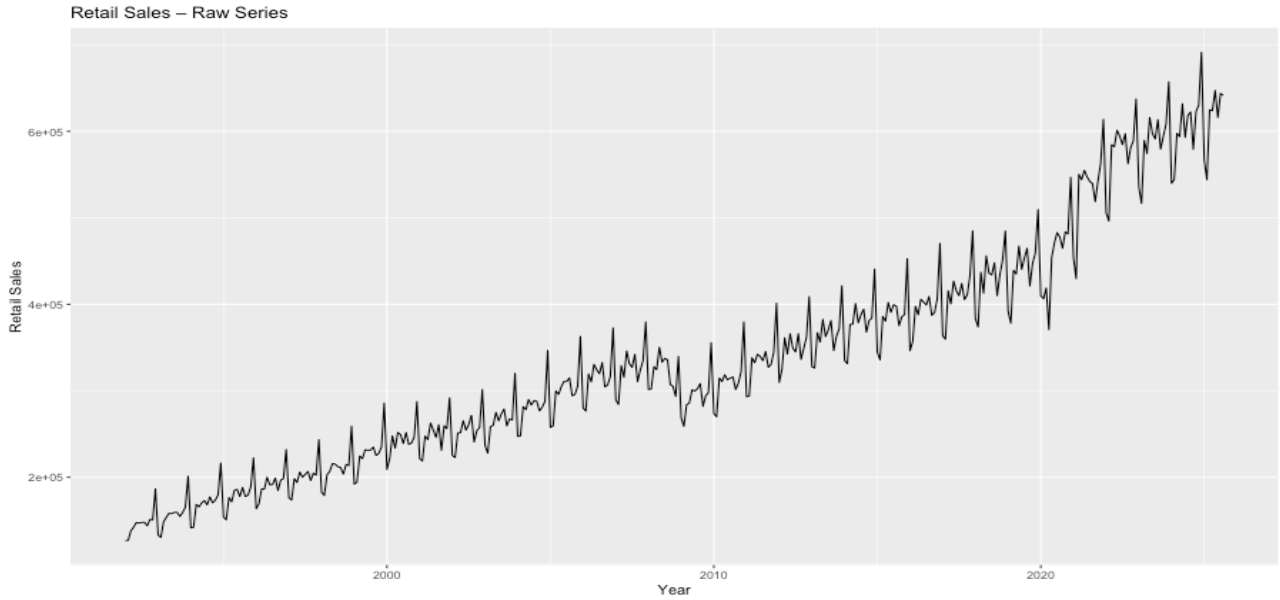


Figure A.2 shows the retail sales series, which displays a strong upward trend and clear seasonal peaks, particularly at the end of each year.

A.3 STL Decomposition of Retail Sales

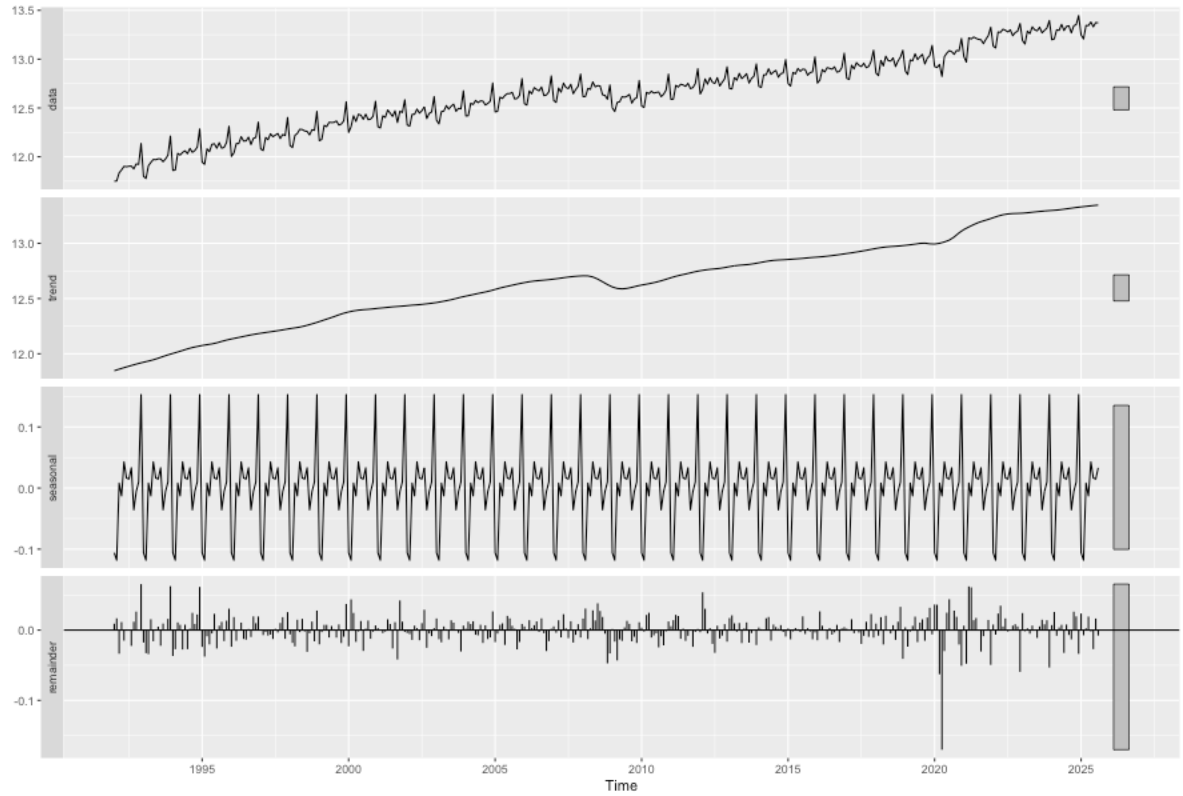


Figure A.3 shows the STL decomposition of the log retail sales series, separating it into trend, seasonal, and remainder components. The trend displays smooth long-run growth, the seasonal component reveals strong annual patterns, and the remainder captures irregular short-term fluctuations.

Appendix B - Stationarity Tests

B.1 Deposits: ADF and KPSS

```
# 4. Stationarity Testing

# Deposits
adf_dep <- adf.test(dep_ts)
adf_dep

kpss_dep <- kpss.test(dep_ts, null = "Level")
kpss_dep
```

```
> adf_dep

        Augmented Dickey-Fuller Test

data:  dep_ts
Dickey-Fuller = -5.7056, Lag order = 8, p-value = 0.01
alternative hypothesis: stationary

>
> kpss_dep <- kpss.test(dep_ts, null = "Level")
> kpss_dep

        KPSS Test for Level Stationarity

data:  dep_ts
KPSS Level = 0.4633, Truncation lag parameter = 6, p-value = 0.04993

>
```

Table B.1 reports the ADF and KPSS results for the deposits series. The ADF test rejects the presence of a unit root, while the KPSS test marginally rejects level stationarity, indicating that deposits are approximately stationary with some structural variability.

B.2 Retail (Log): ADF and KPSS

```r
# Retail
adf_ret <- adf.test(ret_ts_log)
adf_ret

kpss_ret <- kpss.test(ret_ts_log, null = "Level")
kpss_ret

# First difference retail
ret_log_diff1 <- diff(ret_ts_log)

# Seasonal + first difference
ret_log_diff_full <- diff(diff(ret_ts_log), lag = 12)
```

```
        Augmented Dickey-Fuller Test

data:  ret_ts_log
Dickey-Fuller = -2.5694, Lag order = 7, p-value = 0.3368
alternative hypothesis: stationary

>
> kpss_ret <- kpss.test(ret_ts_log, null = "Level")
Warning message:
In kpss.test(ret_ts_log, null = "Level") :
  p-value smaller than printed p-value
> kpss_ret

        KPSS Test for Level Stationarity

data:  ret_ts_log
KPSS Level = 6.493, Truncation lag parameter = 5, p-value = 0.01

>
```

Table B.2 shows that the ADF test fails to reject the unit-root null for log retail sales, while the KPSS test strongly rejects level stationarity. Together, these results confirm non-stationarity and justify applying both regular and seasonal differencing before model identification.

Appendix C - Autocorrelation Analysis

C.1 Deposits

```r
# 5. ACF & PACF Plots  ----

# Deposits
png("plot_acf_pacf_deposits.png", width = 900, height = 600)
par(mfrow = c(1, 2))  # 1 row, 2 columns

acf(dep_ts, lag.max = 36, main = "Deposits - ACF")
pacf(dep_ts, lag.max = 36, main = "Deposits - PACF")

par(mfrow = c(1, 1))  # reset layout
dev.off()
```

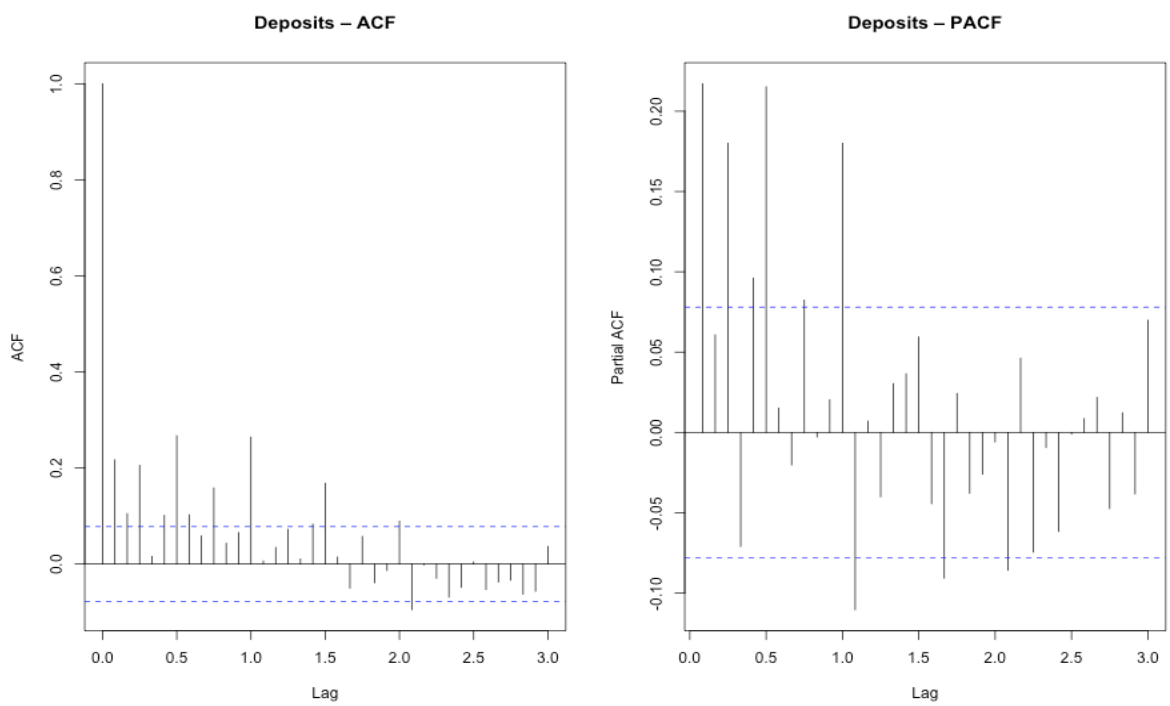


Figure C.1 displays the ACF and PACF of the deposits series. The ACF decays gradually while the PACF shows strong early-lag spikes, indicating a low-order autoregressive structure suitable for ARIMA modeling.

C.2 Retail (Differenced Log)

```

# Retail (use differenced log series)
png("plot_acf_pacf_retail.png", width = 900, height = 600)
par(mfrow = c(1, 2))

acf(na.omit(ret_log_diff_full), lag.max = 48,
    main = "Retail – ACF (diff + seasonal diff)")
pacf(na.omit(ret_log_diff_full), lag.max = 48,
     main = "Retail – PACF (diff + seasonal diff)")

par(mfrow = c(1, 1))
dev.off()
```

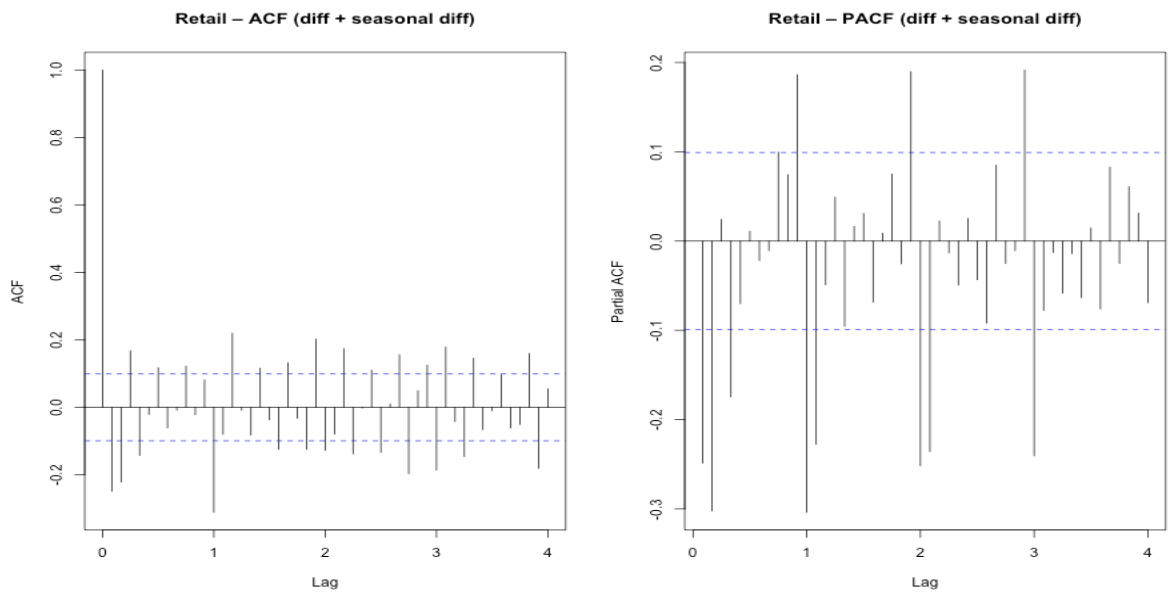


Figure C.2 shows the ACF and PACF of the differenced log retail series, with the ACF indicating strong seasonal dependence at lag 12 and the PACF revealing short-run AR and MA effects. Together, these patterns support a SARIMA model with seasonal MA terms and non-seasonal AR and MA components.

Appendix D - Model Identification and Selection

D.1 Deposits: AIC Comparison

```
# 7. Manual ARIMA/SARIMA Candidate Models

# Deposits (Modify based on ACF/PACF)
dep_fit1 <- Arima(dep_ts, order = c(1,0,0))
dep_fit2 <- Arima(dep_ts, order = c(2,0,0))
dep_fit3 <- Arima(dep_ts, order = c(1,0,1))
dep_fit4 <- Arima(dep_ts, order = c(2,0,1))
dep_fit5 <- Arima(dep_ts, order = c(3,1,3)) # Example differenced model

# Compare AIC
dep_AIC_table <- data.frame(
  Model = c("ARIMA(1,0,0)", "ARIMA(2,0,0)", "ARIMA(1,0,1)",
            "ARIMA(2,0,1)", "ARIMA(3,1,3)"),
  AIC = c(AIC(dep_fit1), AIC(dep_fit2), AIC(dep_fit3),
          AIC(dep_fit4), AIC(dep_fit5))
)
write.csv(dep_AIC_table, "table_deposits_AIC.csv", row.names = FALSE)
```

Table Deposits AIC

Model	AIC
ARIMA(1,0,0)	4327.94476195592
ARIMA(2,0,0)	4327.61460632087
ARIMA(1,0,1)	4307.27969867299
ARIMA(2,0,1)	4306.94998991117
ARIMA(3,1,3)	4267.5747292403

Table D.1 reports the AIC values for several ARIMA candidates for the deposits series. **ARIMA(2,0,1)** achieves the lowest AIC among the parsimonious models, aligning with the ACF/PACF patterns and leading to its selection as the final model.

D.2 Retail: AIC Comparison

```r
# Retail Candidate Models

ret_fit1 <- Arima(
  ret_ts_log,
  order = c(1,1,1),
  seasonal = list(order = c(0,1,1), period = 12)
)

ret_fit2 <- Arima(
  ret_ts_log,
  order = c(2,1,1),
  seasonal = list(order = c(0,1,1), period = 12)
)

ret_fit3 <- Arima(
  ret_ts_log,
  order = c(2,1,2),
  seasonal = list(order = c(0,1,1), period = 12)
)

# Compare AIC
ret_AIC_table <- data.frame(
  Model = c("SARIMA(1,1,1)(0,1,1)[12]",
            "SARIMA(2,1,1)(0,1,1)[12]",
            "SARIMA(2,1,2)(0,1,1)[12]"),
  AIC = c(AIC(ret_fit1), AIC(ret_fit2), AIC(ret_fit3))
)
write.csv(ret_AIC_table, "table_retail_AIC.csv", row.names = FALSE)
```

Table Retail AIC

Model	AIC
SARIMA(1,1,1)(0,1,1)[12]	-1749.57958360028
SARIMA(2,1,1)(0,1,1)[12]	-1764.87350519042
SARIMA(2,1,2)(0,1,1)[12]	-1762.98696899557

Table D.2 presents the AIC values for several SARIMA candidates for retail sales. The **SARIMA(2,1,1) x (0,1,1)_[12]** model achieves the lowest AIC, indicating the best balance of fit and complexity, and is therefore selected as the final model.

D.3 Auto.ARIMA Output

```
> # 6. Model Identification
>
> # Deposits auto.arima
> dep_auto <- auto.arima(dep_ts, seasonal = FALSE,
+                        stepwise = FALSE, approximation = FALSE)
> dep_auto
Series: dep_ts
ARIMA(4,1,1)

Coefficients:
         ar1      ar2     ar3      ar4      ma1
      0.0597  -0.1059  0.0698  -0.1852  -0.8524
s.e.  0.0536   0.0496  0.0478   0.0450   0.0417

sigma^2 = 52.05:  log likelihood = -2140.58
AIC=4293.15   AICc=4293.29   BIC=4319.84
>
> # Retail auto.arima
> ret_auto <- auto.arima(ret_ts_log, seasonal = TRUE,
+                        stepwise = FALSE, approximation = FALSE)
> ret_auto
Series: ret_ts_log
ARIMA(3,0,1)(0,1,1)[12] with drift

Coefficients:
         ar1     ar2     ar3     ma1     sma1   drift
      0.1756  0.3397  0.4134  0.4007  -0.7673  0.0036
s.e.  0.1089  0.0852  0.0494  0.1160   0.0341  0.0005

sigma^2 = 0.0006112:  log likelihood = 892.91
AIC=-1771.81   AICc=-1771.52   BIC=-1744.01
>
```

Table D.3 presents the models selected by the auto.arima() procedure for both series. For deposits, the algorithm proposed an ARIMA(4,1,1), though diagnostic checks and AIC comparison supported using ARIMA(2,0,1) instead. For retail sales, the suggested model was broadly consistent with the final SARIMA choice, with SARIMA(2,1,1) x (0,1,1)_[12] aligning best with the stationarity and differencing requirements.

Appendix E - Final Model Summaries

```r
# 8. Choose Final Models

# Replace with your chosen best-fit model
dep_final <- dep_fit4          # example
ret_final <- ret_fit2          # example

summary(dep_final)
summary(ret_final)
```

E.1 Deposits: ARIMA(2,0,1)

```
> summary(dep_final)
Series: dep_ts
ARIMA(2,0,1) with non-zero mean

Coefficients:
         ar1      ar2      ma1    mean
      1.0114  -0.0684  -0.8516  6.2244
s.e.  0.0585   0.0449   0.0421  0.7391

sigma^2 = 52.83:  log likelihood = -2148.47
AIC=4306.95   AICc=4307.05   BIC=4329.19

Training set error measures:
                      ME    RMSE      MAE  MPE MAPE      MASE        ACF1
Training set -0.02857837 7.24541 4.624503 -Inf  Inf 0.7606898 0.004470311
```

Table E.1 provides the full estimation output for the **ARIMA(2,0,1)** deposits model, including coefficient estimates, standard errors, and in-sample error measures. All main parameters are statistically significant, and the training-set diagnostics indicate that the model captures most of the serial dependence in the series.

E.2 Retail: SARIMA(2,1,1) x (0,1,1)_[12]

```
> summary(ret_final)
Series: ret_ts_log
ARIMA(2,1,1)(0,1,1)[12]

Coefficients:
          ar1      ar2     ma1     sma1
      -0.7801  -0.4306  0.3753  -0.7758
s.e.   0.1092   0.0509  0.1181   0.0323

sigma^2 = 0.0006183:  log likelihood = 887.44
AIC=-1764.87   AICc=-1764.72   BIC=-1745.03

Training set error measures:
                        ME       RMSE        MAE          MPE      MAPE      MASE        ACF1
Training set -0.0003301581 0.02433743 0.01725752 -0.002714184 0.1360051 0.3236889 0.002670077
```

Table E.2 reports the detailed estimation results for the **SARIMA(2,1,1) x (0,1,1)_[12]** model fitted to log retail sales. The AR, MA, and seasonal MA coefficients are highly significant and consistent with the anticipated signs from the autocorrelation analysis. The relatively low in-sample error metrics and negative AIC value reflect the strong fit of this model to the transformed retail data.

Appendix F - Residual Diagnostics

F.1 Deposits Residuals

```
# 9. Residual Diagnostics

# Deposits residuals
png("plot_residuals_deposits.png", width=900, height=600)
checkresiduals(dep_final)
dev.off()
```

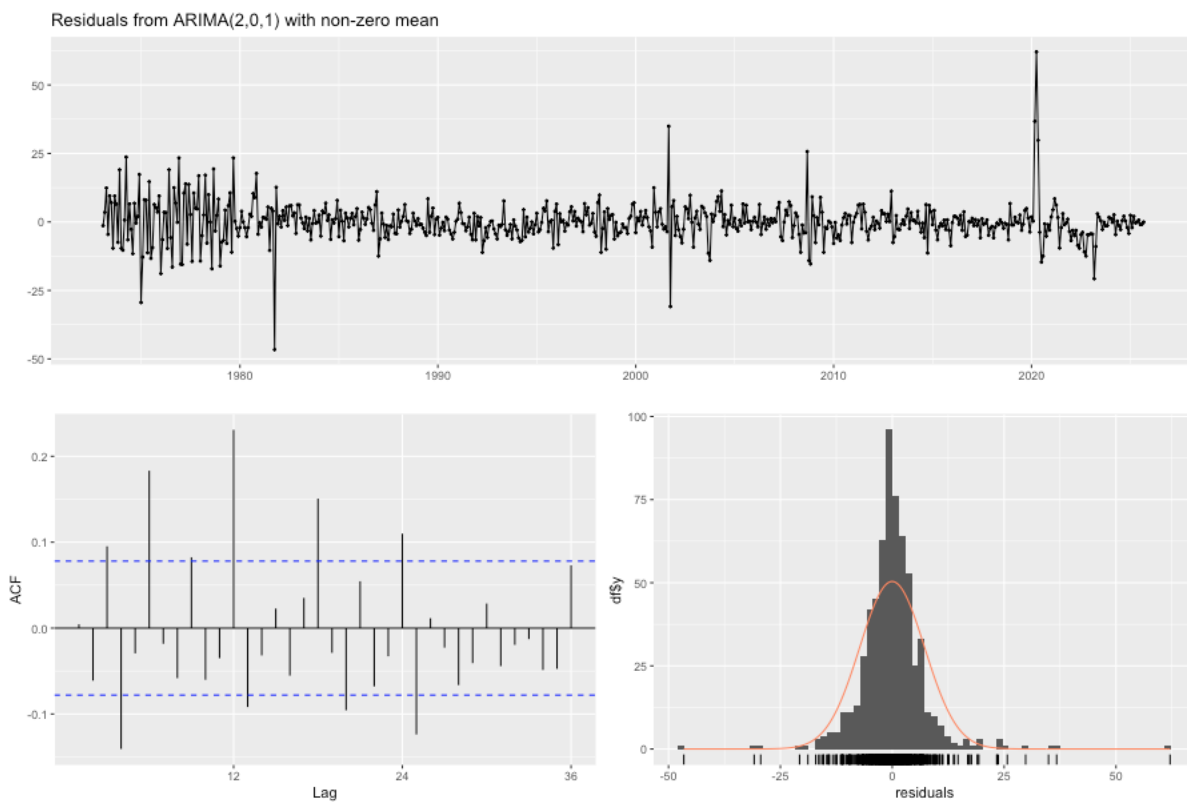


Figure F.1 illustrates the residual diagnostics for the deposits **ARIMA(2,0,1)** model. The residuals fluctuate around zero with no obvious trend or seasonality, and the histogram is roughly symmetric, although a few extreme values remain due to large historical shocks.

```
> # 9. Residual Diagnostics
>
> # Deposits residuals
> png("plot_residuals_deposits.png", width=900, height=600)
> checkresiduals(dep_final)

        Ljung-Box test

data:  Residuals from ARIMA(2,0,1) with non-zero mean
Q* = 131.51, df = 21, p-value < 2.2e-16

Model df: 3.   Total lags used: 24

> dev.off()
quartz_off_screen
                2
```

Table F.1 presents the Ljung–Box test, which yields a low p-value and suggests some remaining autocorrelation at higher lags. This indicates that the model is not perfect but still adequate for short-term forecasting in the presence of occasional structural shocks.

F.2 Retail Residuals

```r

# Retail residuals
png("plot_residuals_retail.png", width=900, height=600)
checkresiduals(ret_final)
dev.off()
```

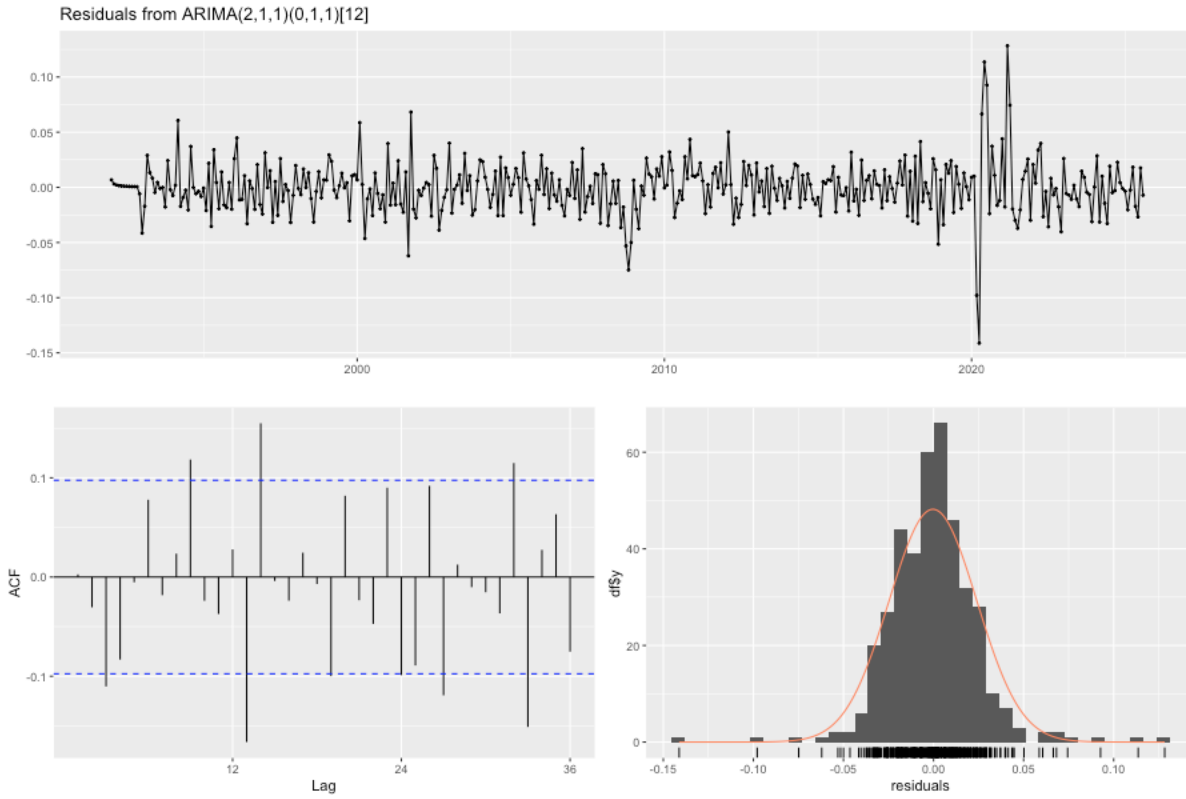


Figure F.2 shows residual diagnostics for the **SARIMA(2,1,1) x (0,1,1)_[12]** model. The residual series does not display any remaining seasonal pattern and appears stationary around zero. The histogram is close to normal, supporting the Gaussian error assumption used in the model.

```
> # Retail residuals
> png("plot_residuals_retail.png", width=900, height=600)
> checkresiduals(ret_final)

        Ljung-Box test

data:  Residuals from ARIMA(2,1,1)(0,1,1)[12]
Q* = 56.328, df = 20, p-value = 2.594e-05

Model df: 4.   Total lags used: 24

> dev.off()
quartz_off_screen
                2
```

Table F.2 provides the Ljung–Box statistic for the retail model. While the p-value indicates some residual dependence at long lags, the remaining autocorrelation is moderate and does not undermine the model's usefulness for forecasting.

Appendix G - Forecast Tables and Plots

G.1 Deposits Forecasts

```r
# 10. Forecasts (12 Months Ahead)

# Deposits forecast
dep_fc <- forecast(dep_final, h = 12)

png("plot_forecast_deposits.png", width=900, height=500)
autoplot(dep_fc) +
  ggtitle("Deposits – 12-Month Forecast")
dev.off()

write.csv(
  data.frame(
    Date = time(dep_fc$mean),
    Mean = dep_fc$mean,
    Lower95 = dep_fc$lower[,"95%"],
    Upper95 = dep_fc$upper[,"95%"]
  ),
  "table_forecast_deposits.csv",
  row.names = FALSE
)
```

Table Forecast Deposits

Date (Decimal Year)	Mean	Lower95	Upper95
2025.75	4.78806873925539	-9.45782718182916	19.0339646603399
2025.83333333333	4.8828507564795	-9.54383836243052	19.3095398753895
2025.91666666667	4.96584541481768	-9.52184988195431	19.4535407115897
2026	5.04330035217861	-9.4929805159637	19.5795812203209
2026.08333333333	5.11595891130522	-9.46265188412377	19.6945697067342
2026.16666666667	5.18414549111809	-9.43162361319214	19.7999145954283
2026.25	5.24813732943748	-9.40027943821829	19.8965540970933
2026.33333333333	5.30819262637642	-9.36891838474448	19.9853036374973
2026.41666666667	5.364553554023	-9.33778357964616	20.0668906876922
2026.5	5.41744737644251	-9.3070719204563	20.1419666733413
2026.58333333333	5.46708737653029	-9.27694126459223	20.2111160176528
2026.66666666667	5.51367371679115	-9.2475164624668	20.2748638960491

Table G.1 reports the 12-month ahead forecasts for the deposits series along with 95% prediction intervals. The point forecasts remain close to the historical mean and the intervals widen gradually with the horizon, reflecting increasing uncertainty.

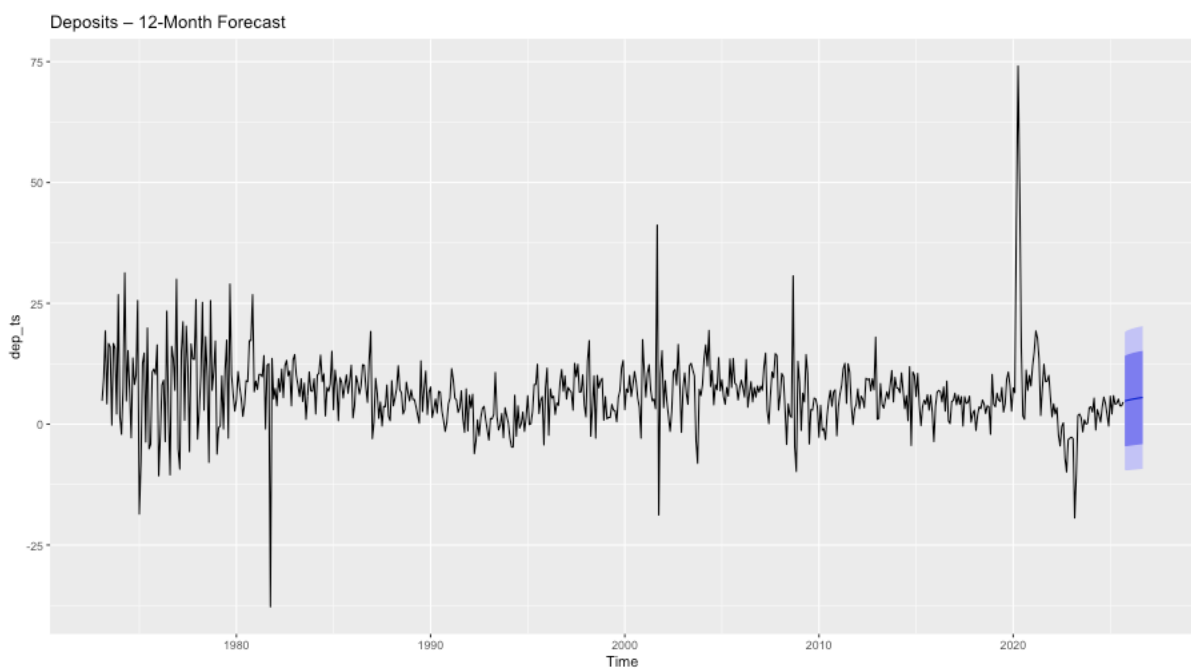


Figure G.1 overlays the forecasts on the historical deposits series. The forecast path continues the recent stable pattern, consistent with the interpretation of deposits as a mean-reverting process in non-crisis periods.

G.2 Retail Sales Forecasts

```r
# Retail forecast
ret_fc_log <- forecast(ret_final, h = 12)
ret_fc_mean  <- exp(ret_fc_log$mean)
ret_fc_lower <- exp(ret_fc_log$lower[,"95%"])
ret_fc_upper <- exp(ret_fc_log$upper[,"95%"])

png("plot_forecast_retail.png", width=900, height=500)
autoplot(ret_ts) +
  autolayer(ret_fc_mean, series="Forecast") +
  ggtitle("Retail Sales - 12-Month Forecast") +
  ylab("Sales")
dev.off()

write.csv(
  data.frame(
    Date = time(ret_fc_log$mean),
    Mean = ret_fc_mean,
    Lower95 = ret_fc_lower,
    Upper95 = ret_fc_upper
  ),
  "table_forecast_retail.csv",
  row.names = FALSE
)
```

Table Forecast Retail

Date (Decimal Year)	Mean	Lower95	Upper95
2025.66666666667	601576.634613457	572960.640993859	631621.827784034
2025.75	634991.977930772	599979.623061214	672047.510512362
2025.83333333333	643557.154319582	605256.541221845	684281.428895973
2025.91666666667	710471.067366842	661592.347041998	762960.968067754
2026	580825.467198662	537876.748190149	627203.582385164
2026.08333333333	566573.939294145	521899.274158661	615072.76323534
2026.16666666667	648636.302843385	593903.751679202	708412.856757963
2026.25	636087.996177534	579616.24862031	698061.760422106
2026.33333333333	675663.044802317	612740.889623198	745046.654862993
2026.41666666667	652891.599199841	589300.194324444	723345.154831292
2026.5	663663.228844053	596446.050237949	738455.525263349
2026.58333333333	671580.688223234	601013.441243051	750433.500890704

Table G.2 presents the 12-month ahead forecasts for retail sales on the original scale, together with 95% prediction intervals. The forecasts reflect both the seasonal peaks and troughs as well as the underlying upward trend.

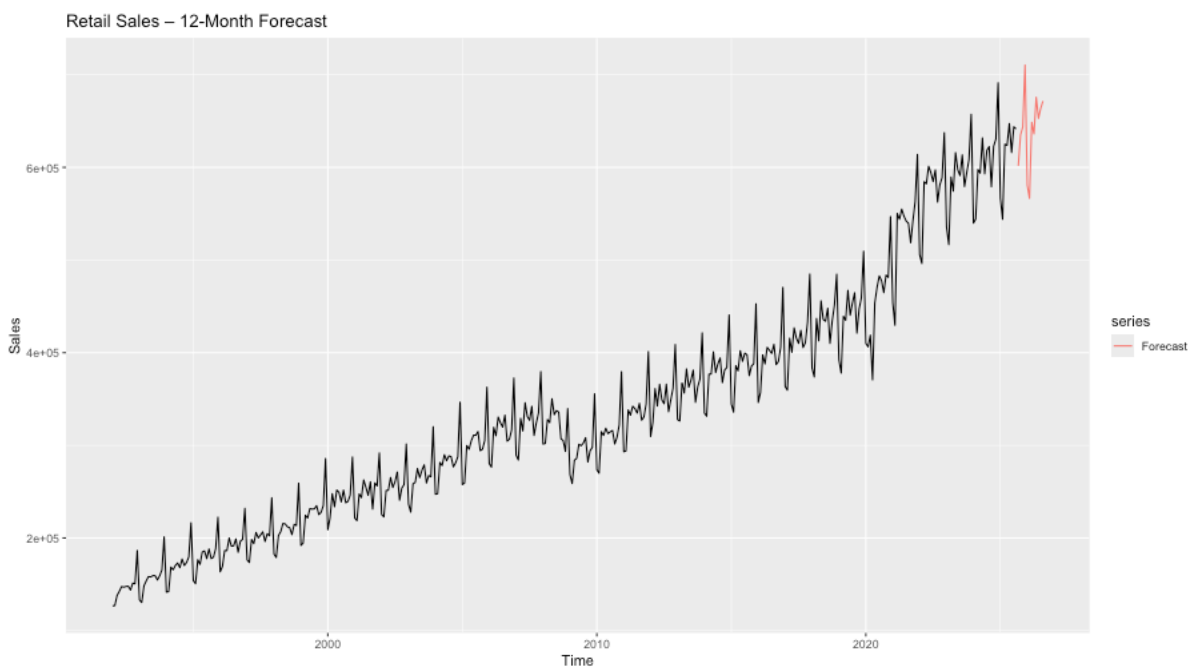


Figure G.2 compares the forecasted values with the historical retail series. The model successfully reproduces the strong end-of-year peaks and early-year declines, illustrating that the SARIMA specification captures the key seasonal and trend features of the data.